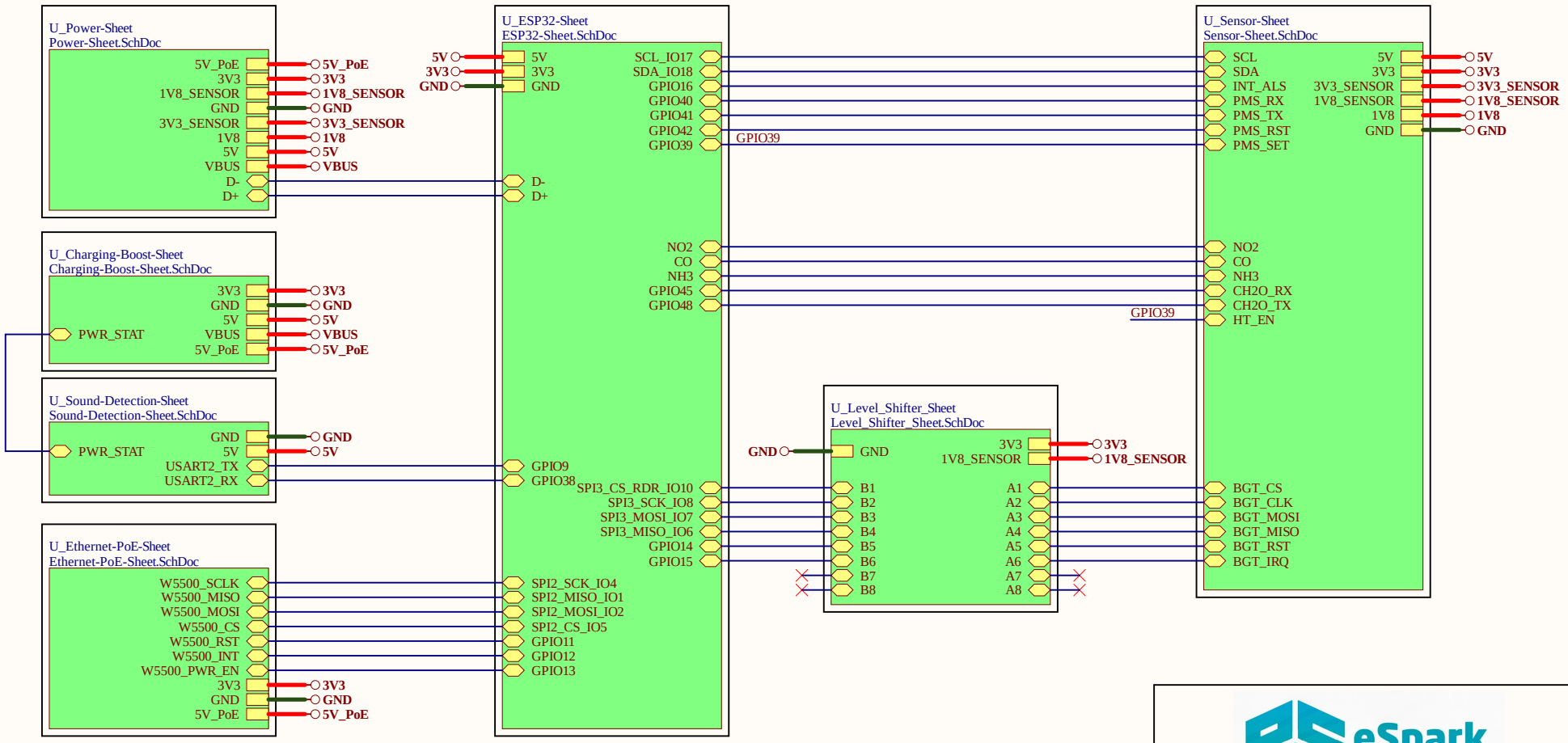
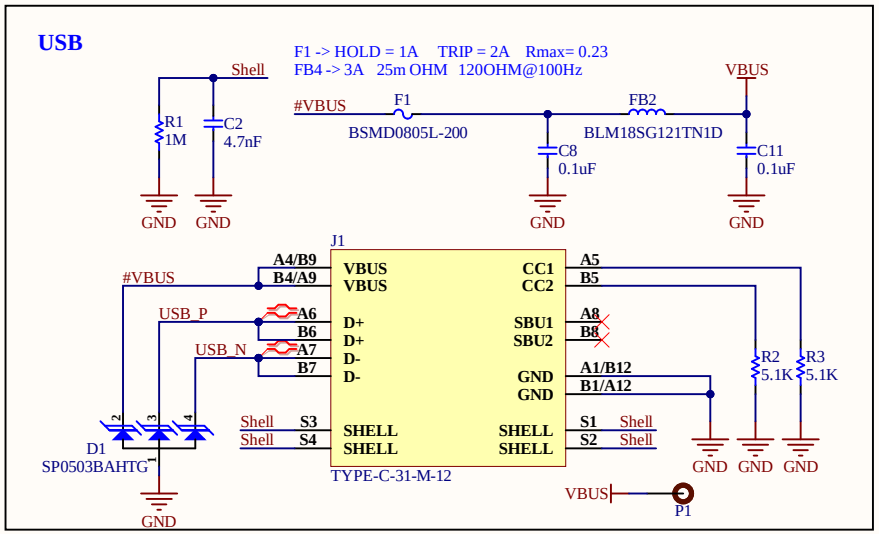
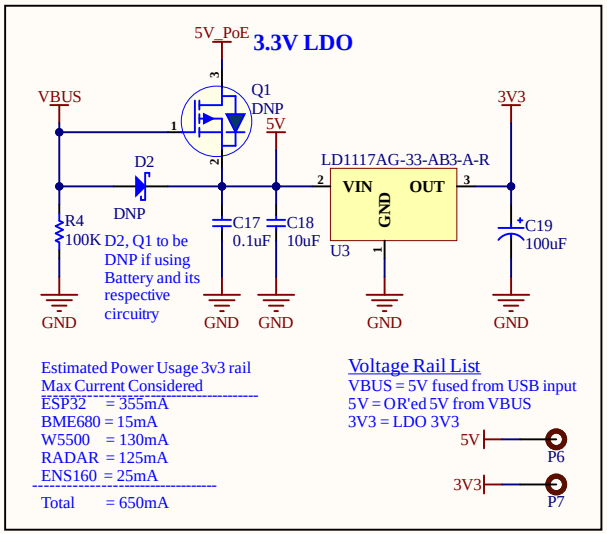
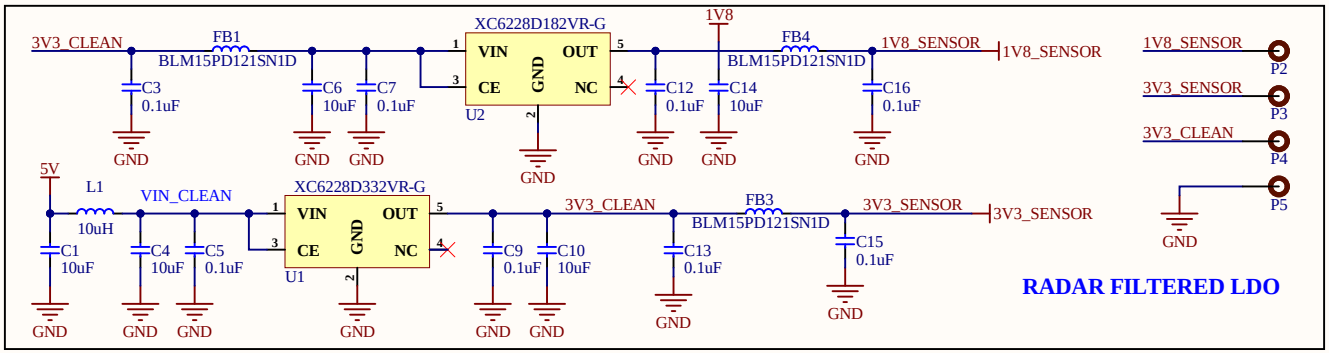


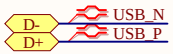
eSpark Consulting Group

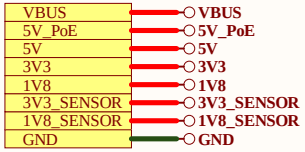
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File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 15-Jun-23	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8







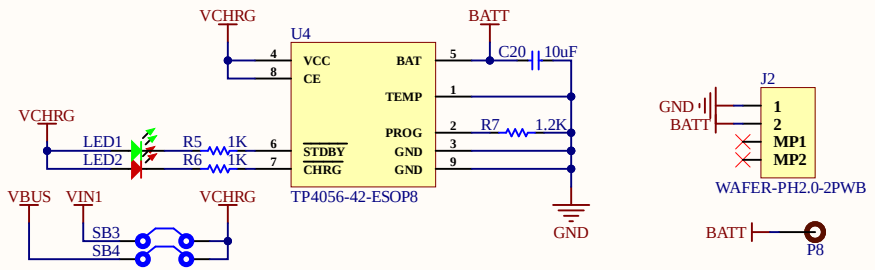




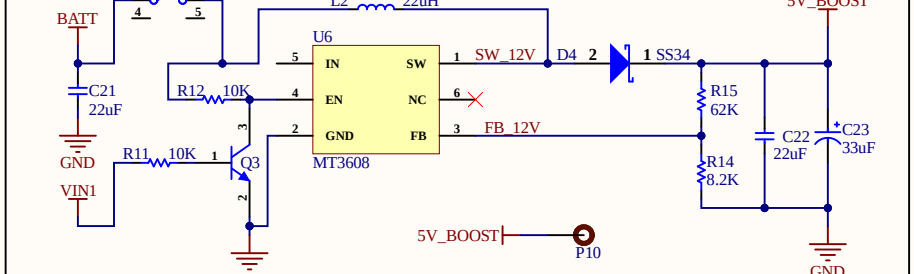
Power Section

Title: Power-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 16-Jun-23	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

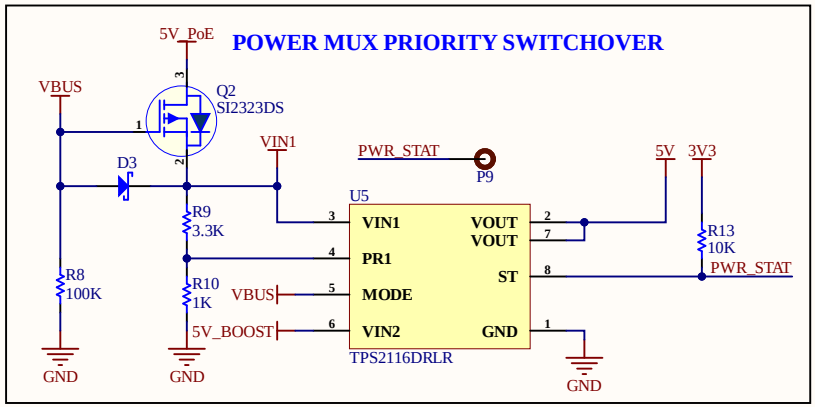
BATTERY CHARGING CIRCUIT



STEP UP CONVERTER



POWER MUX PRIORITY SWITCHOVER



PWR_STAT — PWR_STAT

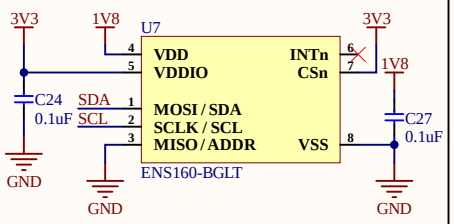
VBUS — VBUS
5V_PoE — 5V_PoE
5V — 5V
3V3 — 3V3
GND — GND

Charging Section

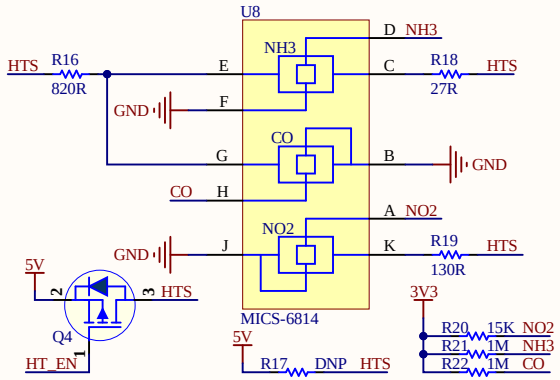
eSpark Consulting Group

Title: Charging-Boost-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 16-Jun-23	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 8

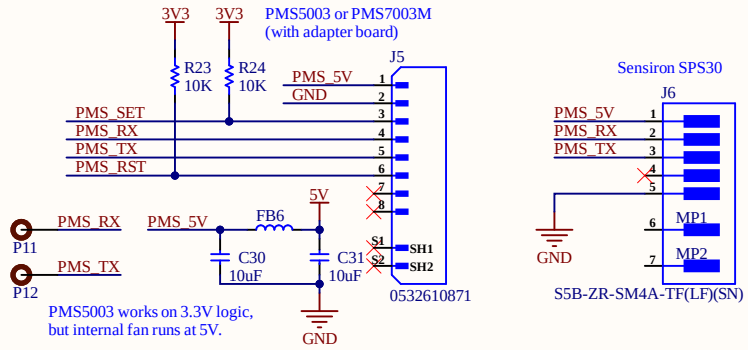
eCO2 TVOC AQI SENSOR



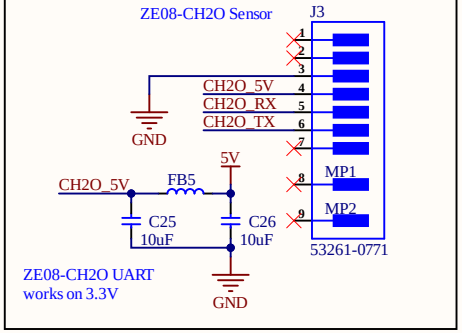
SUPPPORT SENSOR TO DETECT MARIJUANA



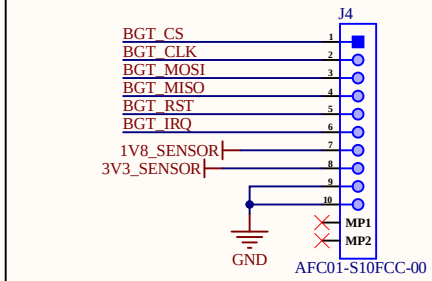
PM2.5 ENVIRONMENT SENSOR CIRCUIT



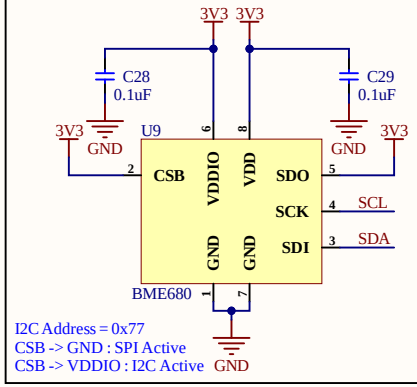
SUPPPORT SENSOR FOR MARIJUANA



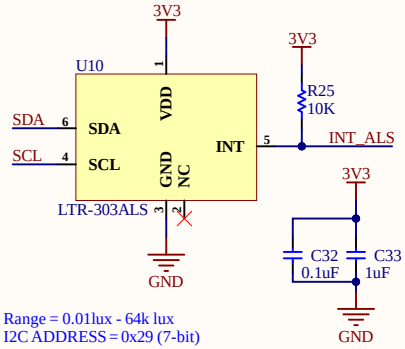
BGT60 RADAR SENSOR

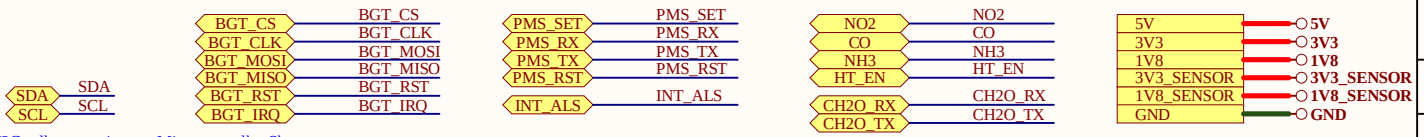


TEMP HUM PRESSURE SENSOR



AMBIENT LIGHT SENSOR CIRCUIT



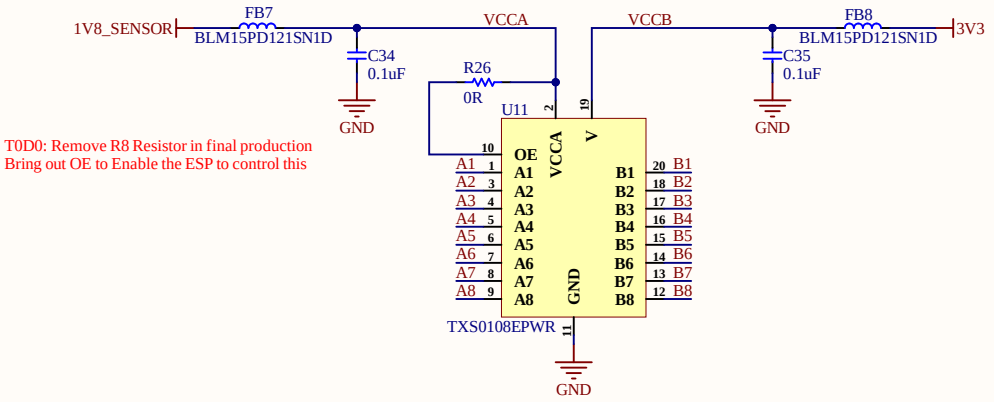


Sensors Section

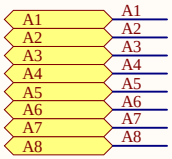
eSpark Consulting Group

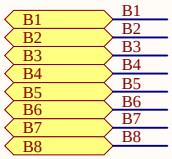
Title: Sensor-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 16-Jun-23	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8

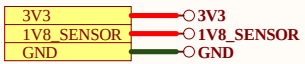
LEVEL SHIFTER



1.4V to 3.6V on A-PORT and 1.65V to 5.5V on B-PORT(VCCA < VCCB)



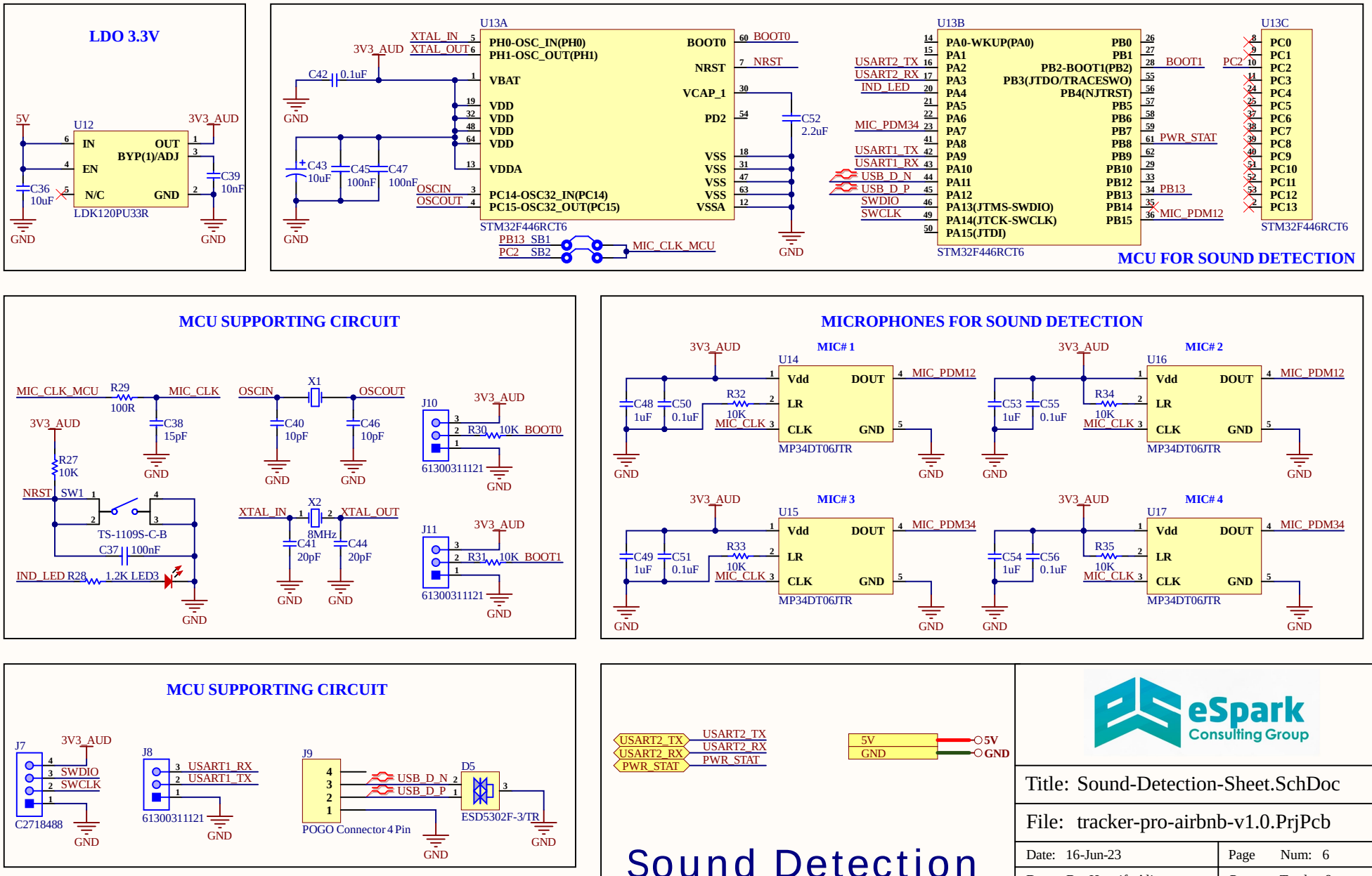


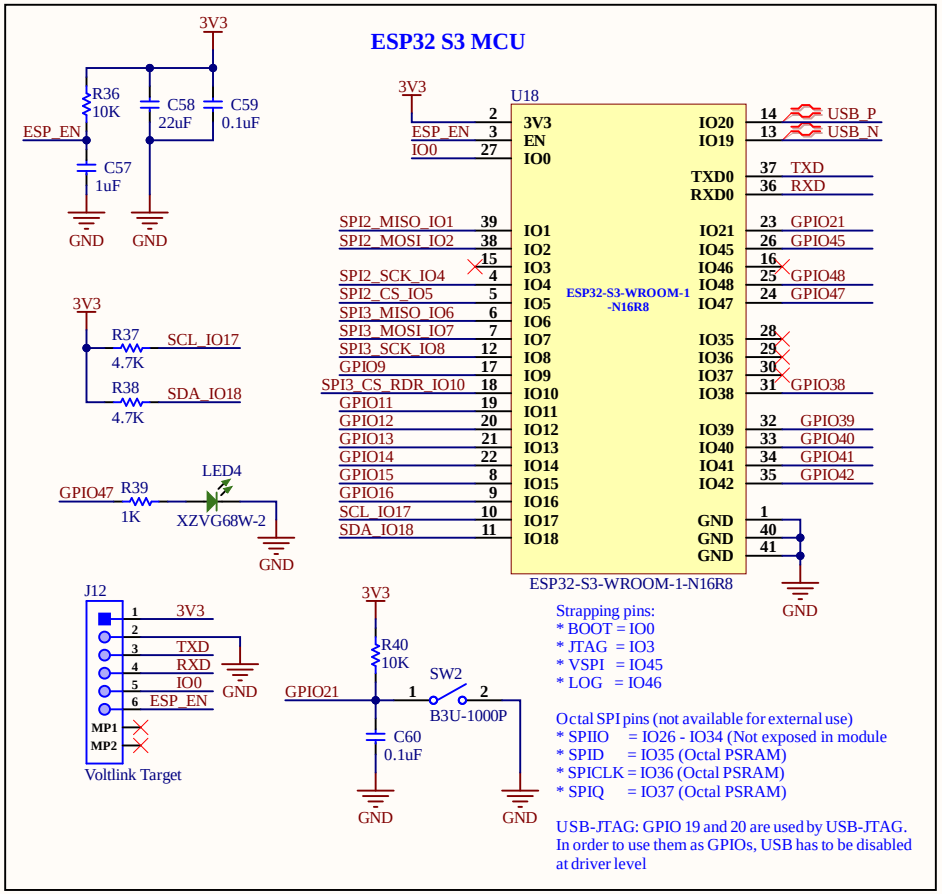


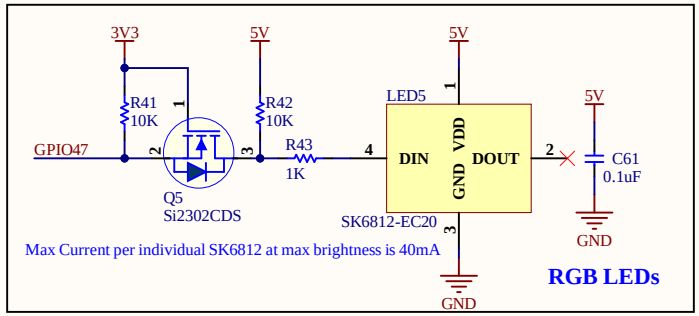
Level Shifter Section

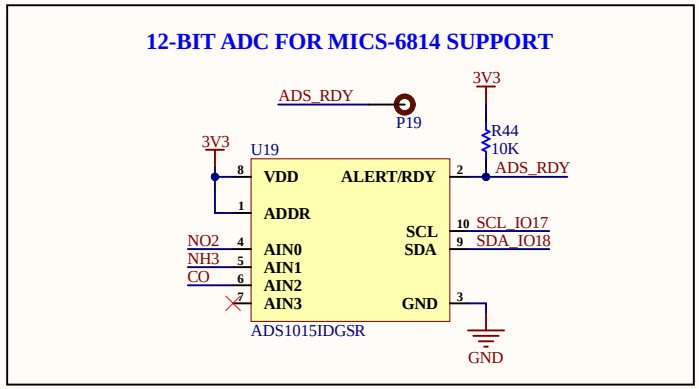
eSpark Consulting Group

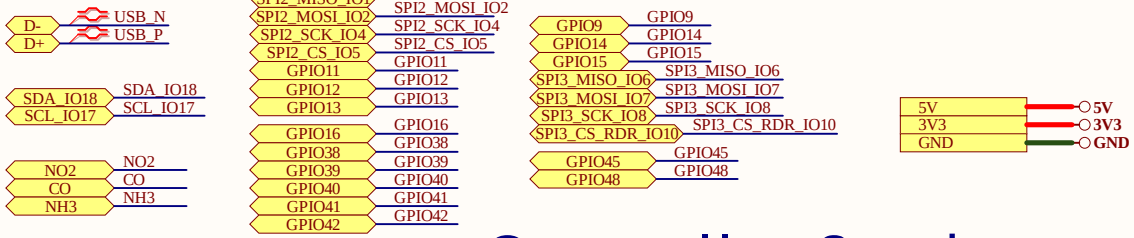
Title: Level_Shifter_Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 16-Jun-23	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


LDO 3.3V
5V
U12
IN
OUT
BYP(1)/ADJ
EN
N/C
GND
LDK120PU33R
3V3_AUD
C36 10uF
C39 10nF
GND
U13A
XTAL_IN
XTAL_OUT
3V3_AUD
C42 0.1uF
PH0-OSC_IN(PH0)
PH1-OSC_OUT(PH1)
VBAT
VDD
VDDA
PC14-OSC32_IN(PC14)
PC15-OSC32_OUT(PC15)
BOOT0
NRST
VCAP_1
PD2
VSS
VSSA
C43 10uF
C45 100nF
C47 100nF
C52 2.2uF
OSCIN
OSCOUT
STM32F446RCT6
PB13 SB1
PC2 SB2
MIC_CLK_MCU
U13B
PA0-WKUP(PA0)
PA1
PA2
PB2-BOOT1(PB2)
PB3(JTDO/TRACESWO)
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
USART2_TX
USART2_RX
IND_LED
MIC_PDM34
USART1_TX
USART1_RX
USB_D_N
USB_D_P
SWDIO
SWCLK
BOOT1
PWR_STAT
PB13
MIC_PDM12
U13C
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
MCU FOR SOUND DETECTION
MCU SUPPORTING CIRCUIT
MIC_CLK_MCU
R29 100R
MIC_CLK
C38 15pF
R27 10K
NRST
SW1
TS-1109S-C-B
C37 100nF
IND_LED
R28
1.2K
LED3
X1
OSCIN
OSCOUT
C40 10pF
C46 10pF
X2
XTAL_IN
XTAL_OUT
8MHz
C41 20pF
C44 20pF
J10
R30 10K
BOOT0
61300311121
J11
R31 10K
BOOT1
MICROPHONES FOR SOUND DETECTION
MIC# 1
MIC# 2
MIC# 3
MIC# 4
U14
U15
U16
U17
Vdd
DOUT
LR
CLK
GND
MP34DT06JTR
MIC_PDM12
MIC_PDM34
C48 1uF
C50 0.1uF
R32 10K
C53 1uF
C55 0.1uF
R34 10K
C49 1uF
C51 0.1uF
R33 10K
C54 1uF
C56 0.1uF
R35 10K
MCU SUPPORTING CIRCUIT
J7
SWDIO
SWCLK
C2718488
J8
USART1_RX
USART1_TX
61300311121
J9
USB_D_N
USB_D_P
POGO Connector 4 Pin
D5
ESD5302F-3/TR
USART2_TX
USART2_RX
PWR_STAT
5V
GND
Sound Detection
eSpark
Consulting Group
Title: Sound-Detection-Sheet.SchDoc
File: tracker-pro-airbnb-v1.0.PrjPcb
Date: 16-Jun-23
Page Num: 6
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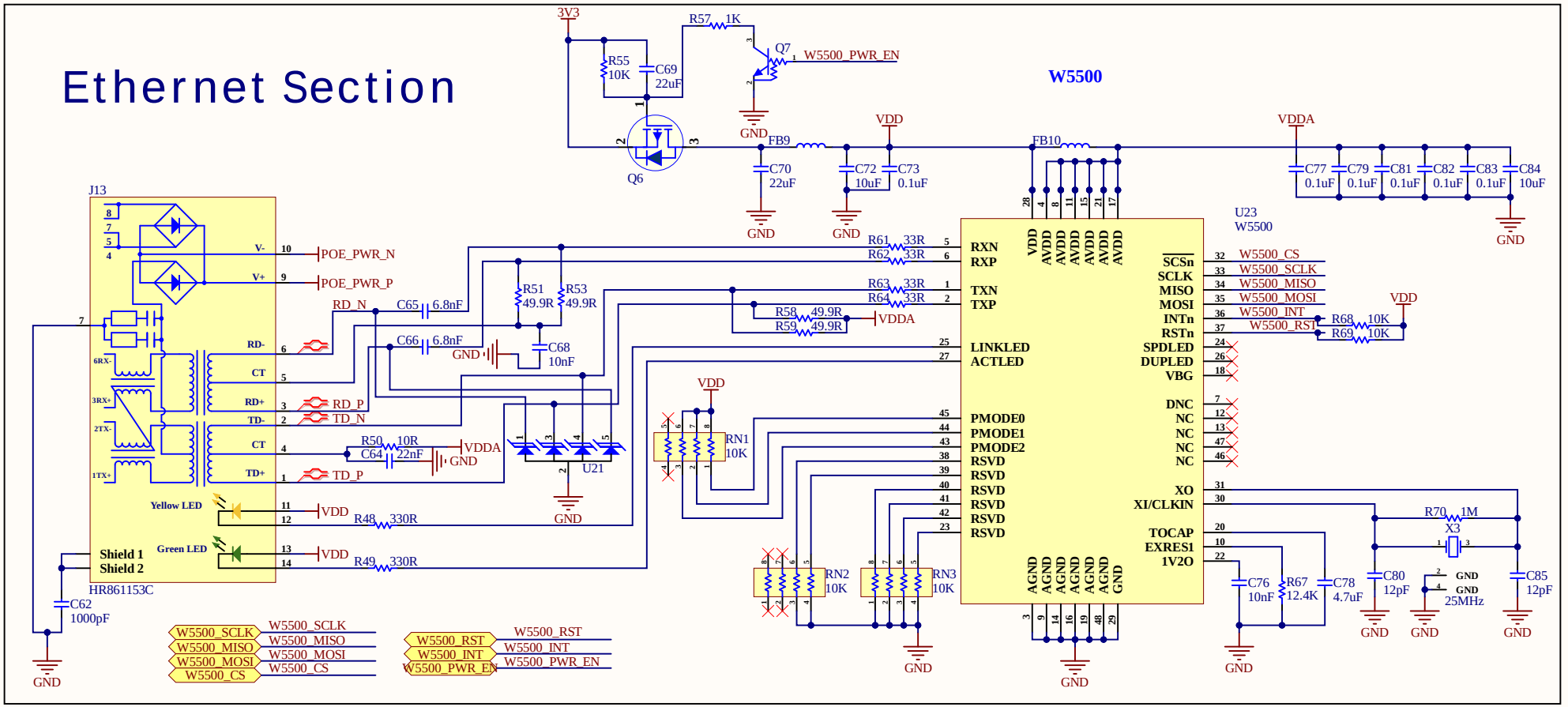






Controller Section

Title: ESP32-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 16-Jun-23	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 8


Ethernet Section
W5500
3V3
R57 1K
R55 10K
C69 22uF
Q7
W5500_PWR_EN
Q6
GND
FB9
VDD
FB10
VDDA
C70 22uF
C72 10uF
C73 0.1uF
C77 0.1uF
C79 0.1uF
C81 0.1uF
C82 0.1uF
C83 0.1uF
C84 10uF
J13
HR861153C
V-
V+
RD-
CT
RD+
TD-
CT
TD+
POE_PWR_N
POE_PWR_P
RD_N
RD_P
TD_N
TD_P
C65 6.8nF
C66 6.8nF
C68 10nF
R51 49.9R
R53 49.9R
R50 10R
C64 22nF
U21
Yellow LED
Green LED
R48 330R
R49 330R
Shield 1
Shield 2
C62 1000pF
U23
W5500
R61 33R
R62 33R
R63 33R
R64 33R
R58 49.9R
R59 49.9R
RXN
RXP
TXN
TXP
LINKLED
ACTLED
PMODE0
PMODE1
PMODE2
RSVD
SCSn
SCLK
MISO
MOSI
INTn
RSTn
SPDLED
DUPLED
VBG
DNC
NC
XO
XI/CLKIN
TOCAP
EXRES1
1V2O
W5500_CS
W5500_SCLK
W5500_MISO
W5500_MOSI
W5500_INT
W5500_RST
R68 10K
R69 10K
RN1 10K
RN2 10K
RN3 10K
R70 1M
X3
25MHz
C76 10nF
R67 12.4K
C78 4.7uF
C80 12pF
C85 12pF
W5500_SCLK
W5500_MISO
W5500_MOSI
W5500_CS
W5500_RST
W5500_INT
W5500_PWR_EN
PoE
POE_PWR_P
POE_PWR_N
D6
C63 0.1uF
R46 24.9K
R45 180K
R47 4.42K
U20
TPS2375PW-1
DET
ILIM
CLASS
VDD
VSS
PG
RTN
NC
C67 15uF
R52 4.7K
R54 30mR
R56 10K
U22
TX4138
VIN
ILIM
ILIM
EN
POK
BS
SW
GND
FB
R60 4.7R
C71 0.1uF
L3 33uH
D7
C74 12pF
R65 301R
R66 57.6K
C75 330uF
F2
D8
5V_PoE
3V3
5V_PoE
GND
Title: Ethernet-PoE-Sheet.SchDoc
File: tracker-pro-airbnb-v1.0.PrjPcb
Date: 16-Jun-23
Page Num: 8
Drawn By: Huzaifa Ali
Page Total: 8